

TABLE II
OTHER NON-INVASIVE TECHNIQUES AND RESULT ANALYSIS

Technique	Principle of determination	Result and analysis
NIR spectroscopy	Absorption or emission data in 0.7-2.5 μ m region was compared to known data for BG. Test sites were ear lobes, finger webs, finger cuticles, forearm skin and lip mucosa (λ -1000nm to 2500nm).	NIR diffuse reflectance performed on fingers and cuticles. Good correlation with BG, 90 per cent of predictions are clinically acceptable.
Raman spectroscopy	Raman spectroscopy measures scattered light that has been influenced by the oscillation and rotation of the scatter. Technique uses laser light to induce molecular emission. Analytes are blood, water, serum and plasma.	Analytical limitations are instability in laser wavelength, intensity, errors due to other chemicals in tissue sample, and long spectral acquisition time.
Photo-acoustic spectroscopy	Photo-acoustic spectroscopy uses an optical beam to rapidly heat the sample and generate an acoustic pressure wave that can be measured by a microphone. Analytes like blood tissue phantoms and humans showed greater sensitivity.	Excellent correlation between photo-acoustic signal and BG levels shown on index fingers. However, instrumentation is currently custom-made, environmentally-sensitive, has chemical and physical interferences and is expensive.
Scatter changes	Light scattering-based measurement monitors the changes in tissue reduced scattering coefficient. Analytes are tissue phantoms and humans. Glucose measurement is proportional to the refractive index of the analyte.	Abdomens of diabetic patients show excellent correlation between scatter signal and BG. But, measurement contributes to natural physiological scattering drift.
IR spectroscopy	Epidermal surface from 10 μ m - 50 μ m enables IR measurements. (λ -700 to 1000nm).	Measurement of glucose in non-blood-containing tissues, such as mucosa. But, interferences due to albumin, red cells, gamma globulin, pH and temperature changes.
Mid-IR spectroscopy (TES)	Thermal emission spectroscopy is the absorption or emission data in 2.5 μ m - 25 μ m region. It is tested to quantify glucose in body fluid.	Non-invasive prototype follows TES, mid-IR spectral region. It measures BG with clinically-acceptable accuracy without individual daily calibrations.

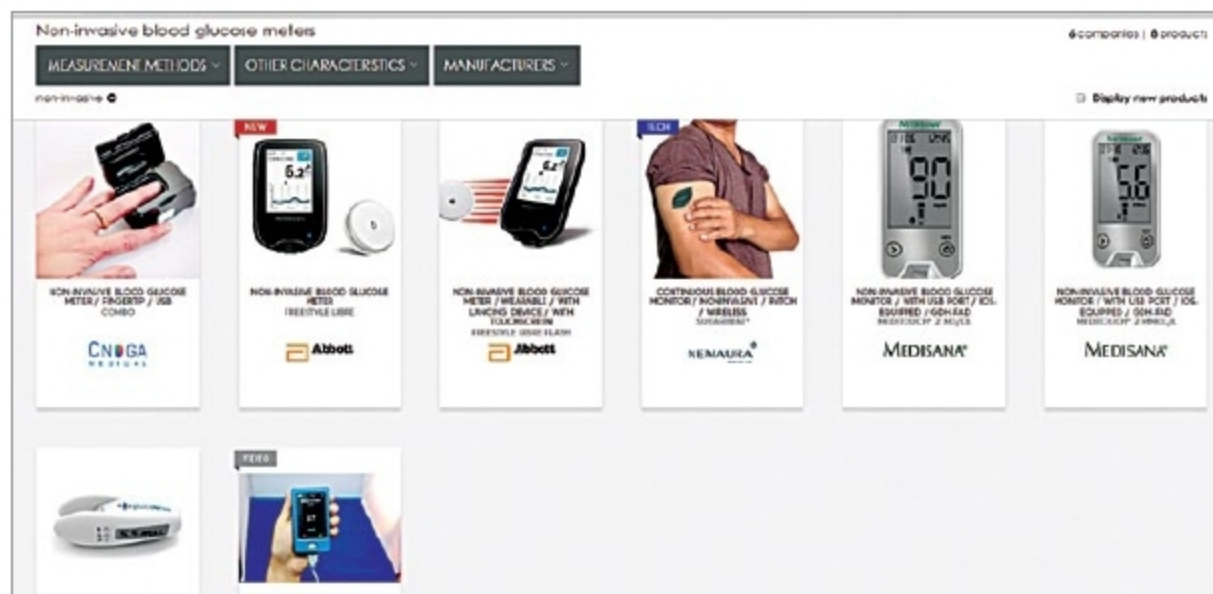


Fig. 11: Some non-invasive glucose meters of the future

non-invasive glucose meters and their usage seem imperative. Soon, indirect contact glucose meters would be feasible, too. Non-invasive BG meters by CNOGA-Combo, Abbott-Freestyle Libre, Nemaura-Sugarbeat, Medisana Glucowise and Glucotrack with continuous monitoring, USB ports, patch/

wireless and touchscreen features are some non-invasive glucose meters of the future (Fig. 11). Many other meters are in their design and development phases. **EFY**

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INSULATION TESTERS

Analog

MC-900BA Series

Digital

DIT 99 Series

Model	Range	Test Voltage DC
MC-901BA	0 - 20 M Ohms	100 V
MC-903BA	0 - 100 M Ohms	500 V
MC-904BA	0 - 500 M Ohms	500 V
MC-901BA	0 - 1000 M Ohms	500 V
MC-906BA	0 - 200 M Ohms	1000 V
MC-907BA	0 - 500 M Ohms	1000 V
MC-901BA	0 - 2000 M Ohms	1000 V

Model	Range	Test Voltage DC	Resolution
DIT 99A	0 - 20 M Ohms	100 V	0.01 M Ohms
DIT 99B	0 - 200 M Ohms	250 V	0.1 M Ohms
DIT 99C	0 - 200 M Ohms	500 V	0.1 M Ohms
DIT 99D	0 - 200 M Ohms	1000 V	0.1 M Ohms
DIT 99E	0 - 2000 M Ohms	1000 V	1 M Ohms

5KV

DIT 954

Specification	Test Voltage	Range
Insulation Resistance	1000V / 2500V / 5000V	0.1MΩ to 200GΩ
AC Voltage Measurement	0 - 600VAC (50 - 60Hz)	-
Phase Sequence Test	100V - 450V (Phase - Phase) 40 - 60Hz	-

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